

## Modal Logic: Tableaux Examples

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2026-02-18

Example 16:  $\Diamond \Box A \rightarrow \Box \Diamond A$  in  $S_4$

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|----|---------------------------------------------------------------|----------------------------|
| 1. | $\mathbb{F} 1, \Diamond \Box A \rightarrow \Box \Diamond A$ ✓ | Assumption                 |
| 2. | $\mathbb{T} 1, \Diamond \Box A$ ✓                             | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, \Box \Diamond A$                               | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{T} 1.1, \Box A$                                      | $\Diamond \mathbb{T} 2$    |

Assume the formula is false at world 1. True  $\Diamond$  (line 2) introduces world 1.1, where  $\Box A$  holds. False  $\Box$  will introduce a second world once we apply it to line 3 — but first let's continue developing world 1.1.

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| 2. | $\mathbb{T} 1, \Diamond \Box A$ ✓                             | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, \Box \Diamond A$ ✓                             | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{T} 1.1, \Box A$                                      | $\Diamond \mathbb{T} 2$    |
| 5. | $\mathbb{F} 1.2, \Diamond A$                                  | $\Box \mathbb{F} 3$        |

False  $\Box$  (line 3) introduces world 1.2, where  $\Diamond A$  is false. We now have two accessible worlds from 1: world 1.1, where  $\Box A$  holds, and world 1.2, where  $\Diamond A$  is false.

|    |                                                               |                            |
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| 2. | $\mathbb{T} 1, \Diamond \Box A$ ✓                             | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, \Box \Diamond A$ ✓                             | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{T} 1.1, \Box A$                                      | $\Diamond \mathbb{T} 2$    |
| 5. | $\mathbb{F} 1.2, \Diamond A$                                  | $\Box \mathbb{F} 3$        |
| 6. | $\mathbb{T} 1.1, A$                                           | $\mathbb{T} \Box 4$        |
| 7. | $\mathbb{F} 1.2, A$                                           | $\mathbb{T} \Diamond 5$    |

The  $\mathbb{T}\Box$  rule gives  $A$  true at 1.1 (line 6), and the  $\mathbb{T}\Diamond$  rule gives  $A$  false at 1.2 (line 7). There is no contradiction: lines 6 and 7 are about **different worlds**, and in S4 there is no rule that connects them. The tree stays open.

## Countermodel for S4

The open branch describes a model:

- Three worlds:  $w_1, w_{1.1}, w_{1.2}$ . World  $w_1$  accesses both  $w_{1.1}$  and  $w_{1.2}$ ; each world accesses itself (T condition). Worlds  $w_{1.1}$  and  $w_{1.2}$  do **not** access each other.
- $A$  is **true** at  $w_{1.1}$  and **false** at  $w_{1.2}$ .
- Since  $w_{1.1}$  only accesses itself and  $A$  is true there,  $\Box A$  is true at  $w_{1.1}$  — so  $\Diamond \Box A$  is true at  $w_1$ .
- Since  $w_{1.2}$  only accesses itself and  $A$  is false there,  $\Diamond A$  is false at  $w_{1.2}$  — so  $\Box \Diamond A$  is false at  $w_1$ .

This is an S4 frame (reflexive and transitive): the only chains of length 2 run through self-loops, so transitivity holds vacuously. But  $w_{1.1}$  and  $w_{1.2}$  are “side by side” — neither can see the other — and that is exactly what S4 cannot rule out.

Example 17:  $\Diamond\Box A \rightarrow \Box\Diamond A$  in S5

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## Example 17: $\Diamond\Box A \rightarrow \Box\Diamond A$ in S5

In **S5** we have the additional **5 rule** (euclidean condition): if two worlds are both accessible from the same world, they can access each other.

So since worlds 1.1 and 1.2 are both accessible from world 1, the 5 rule tells us that 1.1 can access 1.2. Let's pick up where we left off.

|    |                                                               |                            |
|----|---------------------------------------------------------------|----------------------------|
| 1. | $\mathbb{F} 1, \Diamond \Box A \rightarrow \Box \Diamond A$ ✓ | Assumption                 |
| 2. | $\mathbb{T} 1, \Diamond \Box A$ ✓                             | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, \Box \Diamond A$ ✓                             | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{T} 1.1, \Box A$                                      | $\Diamond \mathbb{T} 2$    |
| 5. | $\mathbb{F} 1.2, \Diamond A$                                  | $\Box \mathbb{F} 3$        |
| 6. | $\mathbb{T} 1.1, A$                                           | $\mathbb{T} \Box 4$        |
| 7. | $\mathbb{F} 1.2, A$ ✓                                         | $\mathbb{T} \Diamond 5$    |
| 8. | $\mathbb{T} 1.2, A$                                           | $\Box \mathbb{T} 4$        |



The  $\mathbb{T} \Diamond$  rule gives  $A$  false at 1.2 (line 7, with line 5 now checked). But True  $\Box$  (line 4) is reusable: since the 5 rule gives us that 1.1 accesses 1.2, it fires at world 1.2, giving  $A$  true there. This contradicts line 7. The branch closes. This formula is **valid** in **S5**.

## Why is this valid?

- Assume  $\Diamond\Box A$  at  $w_1$ : some accessible world  $w_{1.1}$  has  $\Box A$ .
- Assume for contradiction that  $\Box\Diamond A$  fails at  $w_1$ : some accessible world  $w_{1.2}$  has  $\Diamond A$  false, meaning  $A$  is false at every world accessible from  $w_{1.2}$ .
- By the T condition,  $w_{1.2}$  accesses itself, so  $A$  is false at  $w_{1.2}$ .
- By the 5 rule (euclidean condition), since  $w_1$  accesses both  $w_{1.1}$  and  $w_{1.2}$ , world  $w_{1.1}$  accesses  $w_{1.2}$ .
- But  $\Box A$  is true at  $w_{1.1}$ , so  $A$  must be true at every world accessible from  $w_{1.1}$  — including  $w_{1.2}$ . Contradiction.

The euclidean condition is essential: it “connects” the two accessible worlds  $w_{1.1}$  and  $w_{1.2}$ , allowing  $\Box A$  at  $w_{1.1}$  to reach  $w_{1.2}$  and override the false  $\Diamond A$  there. Without it, those worlds remain isolated from each other.